MEMS 5701 - Aerospace Propulsion

Air Turbo Rocket Ramjet Project

Due 23 April 2025

1 Introduction

We will use the Air-Augmented Rocket (AAR) and Air-Turbo-Rocket Ramjet (ATRR) propulsion systems to gain practice at designing and analyzing components of an advanced, but simplified propulsion system. These systems are capable of providing thrust over a wide range of Mach numbers, but at the same time are significantly simpler to design and analyze than a turbofan engine. The ATRR system is the ultimate goal for this project. The AAR is a useful step in that process because, while it omits the turbomachinery, it still comprises five of the seven components of the ATRR.

To further simplify these analyses, we will allow the inlet, internal ducts, and nozzle areas to adjust to each flight and thrust condition. Such a "rubber engine" assumption sets each analysis point as a design point. A real engine has far less variable geometry and thus performance compromises must be made for off-design operation.

To help you focus on the thermodynamics and fluid dynamics of the cycle analysis, functions are available to handle bookkeeping, species conservation, and combustion temperature. These small function libraries are available for Excel Visual Basic for Applications (VBA) and for Matlab.

2 Analysis Approach

As we have learned, the key to successfully analyzing the propulsion system is to track the conservation variables:

- Mass, as a single global quantity must be conserved. Mass flow of each atomic species (O, N, and H) must also be conserved. For this implementation, we will not need to track the many, specific molecules into which these atoms arrange themselves (eg. H_2O).
- Enthalpy conservation must include incoming freestream and propellant supply enthalpies, as well as the power lost to the system due to shaft inefficiency. We will use a very approximate method for including the heat of combustion. This is described in the "Tt_burned" section below.
- Using the standard practice we have developed in class, entropy will be tracked based on total pressure, forming the basis of the component efficiencies.
- Momentum will be assessed as impulse function (based on absolute pressure) in the mixer and then as gross thrust and ram drag (based on gauge pressure) for the propulsive force prediction.

2.1 Mass Fractions

We are simplifying the mass composition of air as 23.2% O_2 and 76.8% N_2. Note that the mass fractions of monatomic species are the same. Our molecular weights will be O_2=32, H_2=2, H_2O=18, and N_2=28.164 (which has been adjusted to make up for the fact that we're lumping the contribution of Argon into this one inert species). In the combustor, the hydrogen and oxygen feeds are 100% mass fraction of H_2 and O_2, respectively.

To simplify these analyses, we will hold the ratio of specific heats, γ, to be 1.4 for all mixtures.

The gas constant, R, can be computed from the mixture average molecular weight. In general, the average molecular weight is dependent on temperature because elevated temperatures lead to dissociation of species. An accurate assessment would require a complex equilibrium chemistry calculation, or an even more complex finite rate chemistry calculation.

To keep this project manageable, we will approximate the mixture average molecular weight by assuming that all hydrogen and oxygen atoms in a mixture form as much water as possible. This computation is available in the provided function, MWT_yHyOyN(yH, yO, yN) where the arguments are the mass fractions of hydrogen, oxygen, and nitrogen, atoms respectively. Once MWT is known, R can be computed. From the R and assumed constant γ, Cp can be computed at each station.

2.2 Tt_burned

A combustion efficiency will be applied to approximate the heat of combustion to account for energy that is not released as usable temperature rise due to dissociation. $\eta_b = 1 - a\left(T_{t_burned} - b\right)$ is a simplistic curve fit that produces reasonable trends.

Note that this expression is a simple curve fit to ease calculations on this project and is not applicable to any real system. Correct analyses should use full thermo-chemistry calculations.

Solving for T_{t_burned} produces: $T_{t_burned} = \dfrac{h_{initial} + y_{H2_burned}\, h_{pr}(1+ab)}{C_P + a\, y_{H2_burned}\, h_{pr}}$

where a = 8.8x10^{-5}(/R), b = 540R, and h_{pr} = 1.3x10^9 ft^2/sec^2. $h_{initial}$ is the mass-weighted specific enthalpy of all the flows mixed at the station where T_{t_burned} is being evaluated. This computation is available in the provided function, Ttbrn_yHyOyNhi(yH, yO, yN, hi, g).

This simplistic efficiency model does capture the trend of increased heat release as mixture temperature drops. For example, even for a case where all hydrogen is consumed in the rocket chamber, some additional thermal energy will be released when the rocked flow is mixed with the air in the ram combustor. Physically what is happening is the cooler air mixes with the rocket exhaust, lowering the temperature of the entire mixture. As the temperature drops, some dissociated species such as monatomic O, H, and hydroxyl radicals, OH, combine into water, releasing additional thermal energy to the flow.

A consequence is that each T_{t_burned} calculation is independent of any combustion that may have occurred upstream. For the rocket chamber (station C), the incoming enthalpy, $h_{initial}$ is the mass-weighted sum of the propellant enthalpies. Based on this, the rocket chamber temperature (Tt_C) is computed. The control volume for station 7 is as illustrated in Figure 2. Since Tt_C is completely internal to that control volume its value is ignored for the for the T_{t7} computation. For station 7, $h_{initial}$ includes the mass weighted sum of the propellant enthalpies and the enthalpy of the freestream air. It does not include the heat of combustion released in the rocket chamber. Consistent with this, the mass fraction of hydrogen consumed at station 7 is calculated without regard to any hydrogen consumed in the rocket combustor. One way to think about the independence of the station 7 enthalpy from the rocket chamber reaction is that at station 7 there is no way to differentiate where a given hydrogen molecule connected with a given oxygen atom to release its heat of combustion.

Note that the flow impulse at station 7 is very dependent on what happens in the rocket.

3 The Project

3.1 Basic Assumptions

We will evaluate each propulsion system at two points: A launch point at Mach 0.8 and 30,000 ft altitude; and a cruise point at Mach 4.0 and 60,000 ft altitude. The vehicle powered by this engine requires 4000 lbf of net propulsive force at the launch point and 5000 lbf at the cruise condition. Along with the thrust provided by the main flowpath, we must also account for the propulsion-charged drag on the cowl which decreases our net propulsive force. To assess this drag, we will take the forward projected area of the cowl to be 10% of the inlet stream tube area (A_0). We will assess the drag coefficient (C_D) acting on the area as 0.2 at Mach 0.8 and 0.4 at Mach 4.

The cowl drag is computed as: $D_{cowl} = qC_D A_{cowl} = \frac{1}{2}\rho_0 V_0^2 C_D A_{cowl}$

Assume a "rubber engine" where the inlet area (A_1) matches the freestream capture area (A_0), the inlet recoveries are 0.96 and 0.669 at Mach numbers 0.8 and 4, respectively.

Assume propellant feed temperatures of 540R, chamber pressure of $2x10^5$ psf.

The Mach numbers at stations t and 8 are unity. The section 15 & X to 7 is a constant area mixer, where we assume negligible drag for any flame holders or duct friction and the flow can be assumed isentropic from station 7 through station 9. Station 9 is ideally expanded to ambient pressure with a gross thrust coefficient (Cfg) of 0.96.

3.2 Project Part I: Air-Augmented Rocket (AAR)

For Part I of this project, we will analyze an Air-Augmented Rocket (Figure 1), which omits the turbomachinery of the ATRR. To ease the analysis, a few key parameters will be fixed. The area of the air stream entering the mixer (A_{15}) is set by legislating M_{15} to be 0.2 at flight Mach 4, and 0.1 at flight Mach 0.8. A_X is then set by pressure matching P_X to P_{15}.

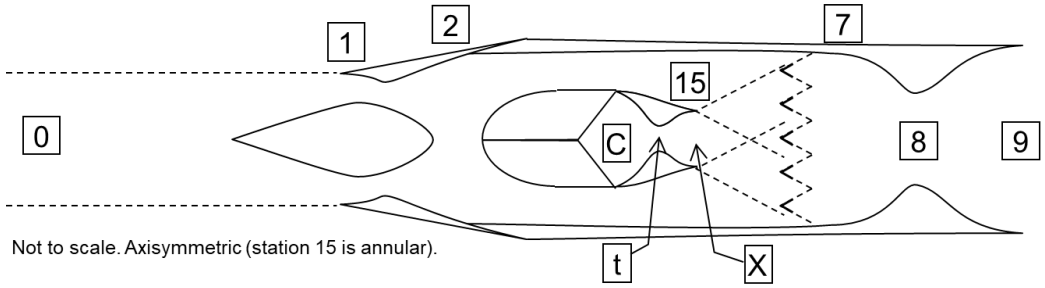


Figure 1: Sketch of Air-Augmented Rocket

For the station 7 calculation, the control volume inflow plane for the impulse function is the rocket exit plane, consisting of stations 15 and X. However, the simplest treatment of the control volume for $h_{initial}$ is the engine as a whole, Figure 2. The bookkeeping for the station 7 T_{t_burned} calculation requires only the enthalpy entering the system from the two propellant flows and the air.

$$h_{initial} = \left(\frac{1}{\dot{m}_7}\right)\left[\dot{m}_{H2}C_{P_{H2}}T_{t_{H2_feed}} + \dot{m}_{O2}C_{P_{O2}}T_{t_{O2_feed}} + \dot{m}_2 C_{P_{air}}T_{t_2}\right]$$

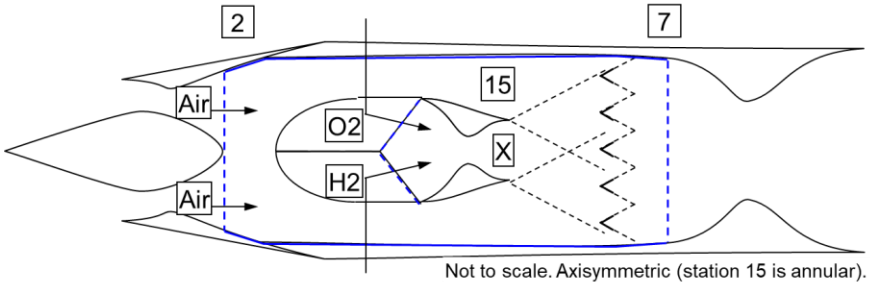


Figure 2: Control Volume for $h_{initial}$ for the AAR

Note that if we were tracking the heats of formation, we would need only the total enthalpy and the constituents at each station. $h_{initial}$ would be embodied in the difference between the total enthalpy and the summed heats of formation. Our simplified approach improves physical insight, avoids the need for full thermo-chemistry evaluations and allows us to use the ideal gas formulations developed in class. The initial enthalpy calculation is the price we pay to be able to use this simplified approach.

You are asked to produce design curves for several combinations of capture area and chamber equivalence ratio. For each flight condition, you will need to iterate on the chamber propellant flow rate to achieve the target thrust. This flowchart may help you form your iteration approach (Figure 3).

Select an A_0 and a chamber equivalence ratio (ϕ)
- Compute $\dot{m}_0 = \dot{m}_{15}$ from A_0.
- Flight condition and recovery determine P_{t15}.
- For the legislated M_{15}=0.2, all properties at 15 are now specified.
- Guess $\dot{m}_C$
  - For the selected ϕ, (and β=0) compute mass flows of H and O.
  - Compute T_{tx}.
  - From $P_X = P_{15}$, compute the properties at the rocket exit.
  - Now all quantities flowing into the mixer are known.
  - Solve for station 7 properties and then gross thrust.
  - Subtract ram drag and cowl drag to get net thrust, F_N.
  - If F_N does not match required for the flight condition, update $\dot{m}_C$.
- If thrust matches, select next A_0 or ϕ.

Figure 3: Air-Augmented Rocket Iteration Scheme

Plot SFC, Isp (including both propellants), specific thrust, and the ratio A9/A0 as the capture area sweeps from near zero (pure rocket) to 15 ft^2 while iterating on the rocket mass flow to match the net propulsive force ($F_g - D_{ram} - D_{cowl}$) to the vehicle requirement. (Note: use $A_0 \sim 1\text{x}10^{-4}$ rather than zero to keep the numerics from becoming undefined.) Do this for rocket equivalence ratios of 1, 2, 10, and 1000. Use good selections for axis limits on your plots to provide visibility into trends.

What is the maximum Isp and how much larger is it than that of the pure rocket? What happens at the Mach 0.8 launch condition and why? Discuss the inlet, mixer, and nozzle exit areas. What happens when these are no longer "rubber"? Document your detailed station properties for the case of ϕ = 10, A_0 = 4.

3.3 Project Part II: Air Turbo-Rocket-Ramjet (ATRR)

For Part II of this project, we will use the same inlet, mixer, combustor, and nozzle assumptions as in Part I, but add the turbomachinery components to the model (Figure 4). After the inlet of the ATRR, the flow is compressed by a fan between stations 12 and 13. The fan is powered by the exhaust of a hydrogen-oxygen rocket that may run fuel-rich. The exhaust from combustion chamber, C, enters the turbine at station 4. The flow is choked downstream of the turbine at t which for this implementation is also the rocket exit, X. The rocket flow properties are set so the exit static pressure, P_X, matches the fan duct pressure, P_{15}. The two streams mix and burn by station 7, then accelerate out of the nozzle, similar to the AAR.

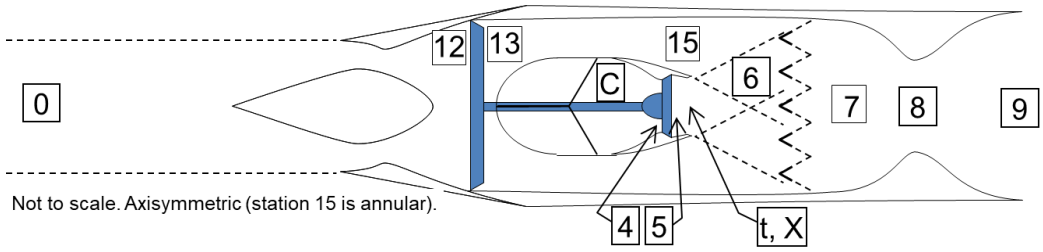


Figure 4: Sketch of Air-Turbo-Rocket Ramjet

The iteration scheme will also change (Figure 5).

Select Π_{fan}
- With flight condition and recovery, this determines P_{T15}.
- For the legislated M_{15}=0.2, P_{15} is now specified.
- $P_X = P_{15}$ and M_X is legislated as 1.0. These define P_{TX}.
- Since P_{TC} is given, we now know $\Pi_{turbine}$.
- Use the definition of $\eta_{turbine}$ to solve for $\tau_{turbine}$ from $\Pi_{turbine}$.
- We now know the specific turbine power: $(\dot{W}/\dot{m})_{turbine} = C_{Pt}(T_{tc} - T_{tx})$.
- Guess A_0
  - With flight condition, this determines $\dot{m}_0$
  - Air flow, η_{fan}, & Π_{fan} determine τ_{fan}.
  - $\dot{W}_{fan} = \dot{m}_0 C_{Pfan}(T_{t13} - T_{t12})$.
  - Since $\dot{W}_{fan}$ must equal $\eta_m \dot{W}_{turbine}$, $\dot{m}_{turbine} = \dot{W}_{fan}/[\eta_m(\dot{W}/\dot{m})_{turbine}]$.
  - Now all quantities flowing into the mixer are known.
  - Solve for station 7 properties and thrust as in Part I.
  - If thrust does not match required for the flight condition guess A_0.
- If thrust matches, select next Π_{fan}.

Figure 5: Air-Turbo-Rocket Ramjet Iteration Scheme

The fan (stations 12 to13) is assumed to have an efficiency η_F of 95% and is driven by the rocket-powered turbine (stations 4 to 5) which has efficiency of 90%. The connecting shaft has efficiency η_M of 99%. For our cases we will hold Mach number at X to be unity (Mx = 1.0). This is not an optimum but it is reasonable and helps limit what would otherwise be a very large design space. We will legislate the Mach number at station 15 to be 0.2 for all flight conditions. We will also force the static pressures between the primary and fan streams at stations X and 15 to match. This forces a relationship between the primary (rocket) and secondary (fan) mass flow rates which simplifies the design space and required iterations.

For $h_{initial}$ in part 2, we need to consider the power transferred by the turbomachinery. One approach is to add the work done by the fan to the enthalpy at station 15 and subtract the work extracted by the turbine from the rocket flow. While this will work, a potentially simpler approach is shown below (Figure 6). By extending the upstream boundary condition to station 2 ahead of the fan, the power transfer between the turbine and the fan is now inside the control volume and irrelevant to the enthalpy sum for station 7… except that the power dissipated by the shaft inefficiency leaves the control volume (sent off to some bearing lubrication cooling system outside our control volume). With this new bookkeeping boundary, the $h_{initial}$ equation is similar to part 1, but with the air conditions evaluated at station 2 rather than 15, and subtracting the shaft inefficiency (the last term).

$$h_{initial} = \left(\frac{1}{\dot{m}_7}\right)\left[\dot{m}_{H2}C_{P\,H2}T_{t_{H2_feed}} + \dot{m}_{O2}C_{P\,O2}T_{t_{O2_feed}} + \dot{m}_2 C_{P\,air}T_{t_2} - \dot{W}_{turbine}(1-\eta_m)\right]$$

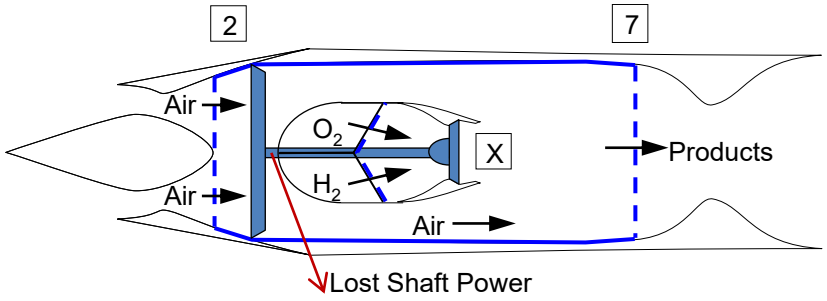


Figure 6: Control Volume for $h_{initial}$ for the ATRR

Plot SFC, Isp (including both propellants), specific thrust, and the ratio A9/A0 as the fan pressure ratio sweeps from 1.1 to 4 while iterating on A_0 to match the net thrust (Fg – Dram – Dcowl) to the vehicle requirement. Do this for rocket equivalence ratios of 1, 2, and 10. Use good selections for axis limits on your plots to provide visibility into trends.

What is the maximum Isp and how much larger is it than that of the pure rocket? What happens at the Mach 0.8 launch condition and why? Discuss the inlet, mixer, and nozzle exit areas. What happens when these are no longer "rubber"? Document your detailed station properties for the launch and cruise cases of FPR = 2, and equivalence ratio of 2. Provide a simple presentation of the results and answers to questions. Aim for 10 to 20 pages including plots and tabulations. I am looking for a brief writeup that demonstrates your understanding of the physics being modeled. A detailed write up is not the focus.